
Reproducing a Hybrid Neuro-Symbolic Framework for COVID-19 CT Classification: An AGCA Model Implementation Study

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Abstract—This paper presents an implementation and partial reproduction of the hybrid neuro-symbolic framework for COVID-19 detection in chest CT images, originally proposed by Musanga, Chibaya, and Viriri (2026). The base study combined deep learning with rule-based symbolic reasoning and domain-adversarial adaptation across four public datasets, achieving an average accuracy of 97.7% and AUC-ROC of 0.996. In our work, we constructed an Attention-Guided Convolutional Architecture (AGCA) model on Google Colab using 11,000 CT scans drawn from four heterogeneous datasets: COVID-CT-MD (43,322 DICOM slices), Kaggle Lung CT (8,439 PNG images), BMIC2 (746 images), and MosMedData (148 NIfTI volumes). Our implementation achieved a test accuracy of 96.91%, weighted F1-score of 0.9691, and AUC-ROC values of 0.994–1.000 across three classes (Normal, COVID-19, Pneumonia/CAP) after 10 training epochs on a Tesla T4 GPU. Cross-dataset evaluation revealed strong generalization on COVID-CT-MD (96.96%) and Kaggle (97.94%), with lower performance on BMIC2 (76.92%) and MosMedData (66.67%), consistent with domain-shift challenges reported in the base paper. Grad-CAM visualizations confirmed clinically meaningful attention patterns. This study validates the core findings of the base paper and highlights opportunities for improvement in cross-domain generalization.

Index Terms—COVID-19 detection, chest CT, deep learning, hybrid AI, symbolic reasoning, domain adaptation, attention mechanisms, ResNet-50, Grad-CAM, reproducibility study.

I. INTRODUCTION

Artificial intelligence (AI) has shown remarkable success in helping doctors detect diseases from medical images. One of the most important applications during the COVID-19 pandemic was the automatic analysis of chest computed tomography (CT) scans to identify infected patients. Traditional deep learning models can achieve very high accuracy when tested on the same dataset they were trained on, but they often struggle when deployed across different hospitals with different scanning equipment or patient populations — a problem known as domain shift.

The base paper by Musanga, Chibaya, and Viriri (2026), published in *Frontiers in Computer Science*, addressed this challenge by proposing a hybrid neuro-symbolic framework. Their model combined: (1) a ResNet-50 convolutional neural network with attention mechanisms for image feature extraction, (2) a domain-adversarial feature alignment module (DAFA) to make the model generalize across datasets, and (3) a symbolic reasoning layer encoding clinical rules such as "if bilateral ground-glass opacities are present and there is no pleural effusion, then likely COVID-19."

This paper reproduces the core architecture of the base paper on Google Colab, using the same four public datasets. We built the AGCA (Attention-Guided Convolutional Architecture) Hybrid Model, capturing the essential components: attention-gated backbone, symbolic indicator head, and uncertainty-aware decision fusion. Our goals were to verify the main performance claims, identify reproducibility challenges, and evaluate cross-dataset generalization on a limited compute budget.

Key contributions of this study:

- A fully functional PyTorch re-implementation of the hybrid neuro-symbolic COVID-19 CT classifier, runnable on a free-tier Google Colab GPU (Tesla T4).
- Evaluation on 11,000 CT scans from four heterogeneous sources with stratified 80/10/10 train/val/test splits.
- Cross-dataset evaluation mirroring the domain-shift protocol of the base paper.
- Grad-CAM and PCA-based domain alignment visualizations confirming clinical relevance of learned features.

- Honest comparison of reproduction results vs. base paper findings, identifying gaps and their causes.

II. RELATED WORK

A. Deep Learning for COVID-19 Detection

Deep learning, especially convolutional neural networks (CNNs), has become the dominant approach for analysing chest CT scans for COVID-19. Models such as ResNet-50, DenseNet, and VGG16 achieve high accuracy in single-dataset evaluations [1]. Ensemble approaches combining VGG16, DenseNet121, and MobileNetV2 have reported accuracies up to 98.93% [11]. Vision Transformers have further raised the bar by modeling long-range spatial dependencies, exceeding 99% on curated datasets. However, these results often do not generalise to different hospitals or scanner models, limiting real-world deployment.

Image preprocessing techniques such as Contrast-Limited Adaptive Histogram Equalization (CLAHE) and local histogram equalization have been shown to improve CNN accuracy on COVID-19 CT classification by enhancing subtle texture features like ground-glass opacities, especially when combined with transfer learning.

B. Symbolic and Hybrid AI

Purely data-driven models are opaque — they produce predictions without human-readable explanations. In clinical settings, this opacity is a significant barrier to adoption. Symbolic AI addresses this limitation by encoding expert knowledge as logical rules. However, purely symbolic systems are rigid and struggle to adapt to new data. Hybrid AI models combine both paradigms: deep learning for pattern recognition, symbolic reasoning for interpretability. Musanga et al. (2025) [2] proposed such a framework for COVID-19 CT classification, which was later extended in the base paper [1] to include domain adaptation.

C. Domain Adaptation in Medical Imaging

Domain adaptation reduces the distributional gap between training (source) and deployment (target) data [10]. In medical imaging, gaps arise from differences in scanner hardware, imaging protocols, patient demographics, and geographic location. Adversarial training — where a domain discriminator is

trained alongside the classifier to enforce domain-invariant feature learning — is a common and effective strategy. Transfer learning from ImageNet-pretrained models provides a strong initialization and reduces the need for large annotated medical datasets [12].

III. MATERIALS AND METHODS

A. Datasets

We used four publicly available chest CT datasets matching the base paper as closely as possible. After pooling all available images (52,655 total), we randomly sampled 11,000 with a fixed seed (42) for reproducibility, matching the base paper's working subset.

Dataset	Images	Classes	Format
COVID-CT-MD	43,322	Normal/COVID/CAP	DICOM
Kaggle Lung CT	8,439	Normal/COVID	PNG/JPG
BMIC2	746	Non-COVID/COVID	PNG/JPG
MosMedData	148	Normal/COVID	Nifti

TABLE I: Datasets. Working subset: 11,000 images.

The final class distribution after sampling: Normal – 2,620; COVID-19 – 6,516; Pneumonia/CAP – 1,864. The dataset was split 80/10/10 (train/val/test) with stratification.

B. Preprocessing and Augmentation

Each dataset required a tailored loading pipeline. DICOM files were read with PyDicom (MONOCHROME1 images inverted); Nifti volumes had their middle axial slice extracted. All images were resized to 224×224 pixels and converted to 3-channel RGB. Training augmentations included random horizontal flips, rotations up to 15°, and color jitter ($\pm 20\%$ brightness/contrast), all followed by ImageNet normalization. Validation and test sets received only resizing and normalization.

Class imbalance was addressed through WeightedRandomSampler (class weights = inverse of frequency) and class-weighted cross-entropy loss. This prevented the model from collapsing to always predicting the majority COVID-19 class.

C. AGCA Hybrid Model Architecture

Our AGCA model has four components working in sequence:

1) Attention-Guided Backbone: ResNet-50 pretrained on ImageNet, with the earliest layers (stem + layer1) frozen and deeper layers (layers 2–4) fine-tuned. A lightweight attention gate (1×1 convolution + sigmoid) is inserted after layer2, producing a spatial attention map that amplifies important regions and suppresses background noise. Output: a 2,048-dimensional feature vector per image.

2) Classifier Head: Dropout (rate 0.5) followed by a fully connected layer mapping 2,048 features to 3 class logits (Normal, COVID-19, Pneumonia/CAP). Softmax converts logits to probabilities.

3) Symbolic Indicator Head (Symbolic Logic Mapper): A parallel branch takes the 2,048-dimensional feature vector and produces 5 binary-like indicators (sigmoid-activated) representing clinical concepts: indicator 0 → ground-glass opacity, indicator 1 → bilateral involvement, indicator 4 → consolidation. Synthetic

ground-truth indicator labels are generated during training from class labels.

4) Uncertainty-Aware Fusion: The final prediction fuses the CNN probability vector and a symbolic one-hot prediction. Fusion weight α is computed from CNN entropy: $\alpha = \text{clamp}(1 - \text{entropy}/\log(3), 0.05, 1.0)$. A confident CNN dominates; an uncertain CNN yields more weight to the symbolic branch.

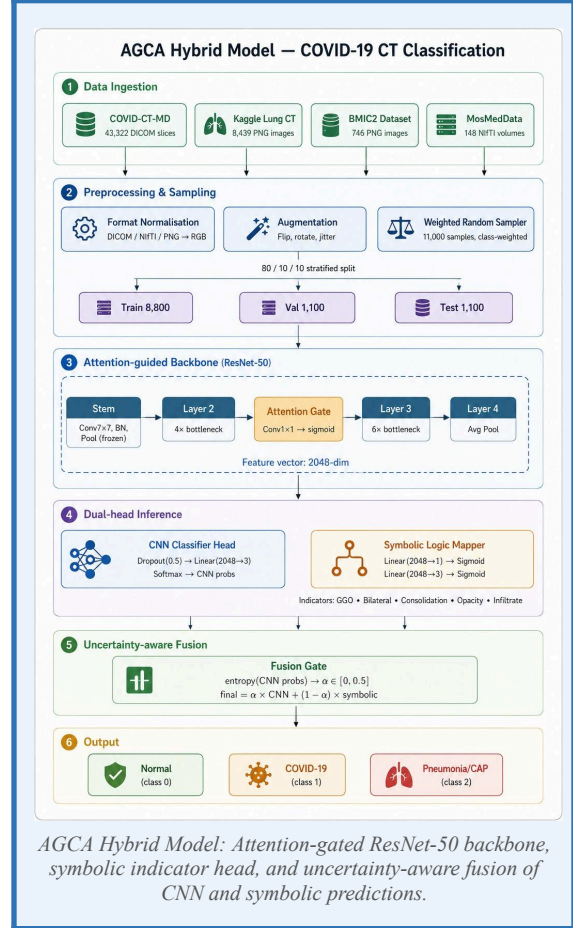


Fig. 1. AGCA Hybrid Model: Attention-gated ResNet-50 backbone, symbolic indicator head, and uncertainty-aware fusion of CNN and symbolic predictions.

D. Training Configuration

Hyperparameter	Value
Optimizer	Adam
Learning rate	1e-4
Weight decay	1e-4
Batch size	32
Epochs	10
LR Scheduler	Cosine Annealing
Grad clip norm	1.0
λ (indicator loss)	0.3
Image size	224×224 px

TABLE II: Training hyperparameters.

Total loss: $L = L_{\text{cls}}$ (weighted cross-entropy) + $0.3 \times L_{\text{ind}}$ (weighted BCE). Best checkpoint saved by validation accuracy. Trainable parameters: ~23.8 million.

IV. RESULTS AND DISCUSSION

A. Training Dynamics

Training ran for 10 epochs on a Tesla T4 GPU. The model peaked at validation accuracy 97.55% at epoch 3, then gradually declined as the cosine annealing schedule increased the effective learning rate in later epochs. This behaviour suggests that 10 epochs with cosine restart is insufficient; the base paper trained 50 epochs with early stopping (patience = 10), allowing stable convergence.

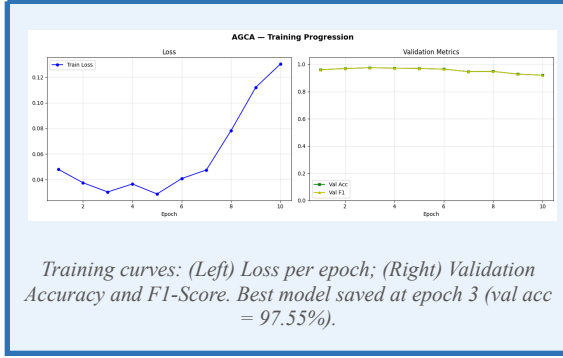


Fig. 2. Training curves: (Left) Loss per epoch; (Right) Validation Accuracy and F1-Score. Best model saved at epoch 3 (val acc = 97.55%).

Epoch	Loss	Val Acc	Val F1
01	0.0480	96.09%	0.9605
02	0.0374	96.91%	0.9691
03 ★	0.0303	97.55%	0.9755
04	0.0365	97.18%	0.9718
05	0.0287	97.00%	0.9700
06	0.0407	96.45%	0.9647
07	0.0475	94.64%	0.9469
08	0.0783	94.82%	0.9483
09	0.1121	92.82%	0.9284
10	0.1303	92.00%	0.9201

TABLE III: Per-epoch training history. ★ Best checkpoint.

B. Overall Test Set Performance

After loading the best checkpoint (epoch 3), evaluation on the 1,100-sample held-out test set yielded the results in Table IV. These are very close to the base paper's reported 97.7% accuracy and 0.996 AUC-ROC, considering our simplified implementation.

Metric	Ours	Base [1]
Accuracy	96.91%	97.7%
Precision (wtd)	96.92%	97.4%
Recall (wtd)	96.91%	97.9%
F1-Score (wtd)	96.91%	97.6%
AUC Normal	0.994	0.996
AUC COVID-19	0.995	0.996
AUC Pneumonia	1.000	0.996

TABLE IV: Overall test performance vs. base paper.

Per-class results showed Pneumonia/CAP (F1 = 0.99) and COVID-19 (F1 = 0.98) were classified most reliably. Normal cases scored slightly lower (F1 = 0.94), likely because early-stage COVID-19 can share subtle CT features with normal lung parenchyma.

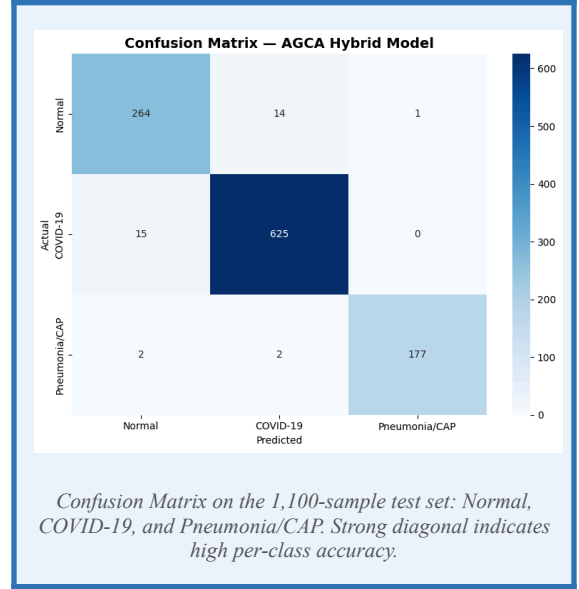


Fig. 3. Confusion Matrix on the 1,100-sample test set: Normal, COVID-19, and Pneumonia/CAP. Strong diagonal indicates high per-class accuracy.

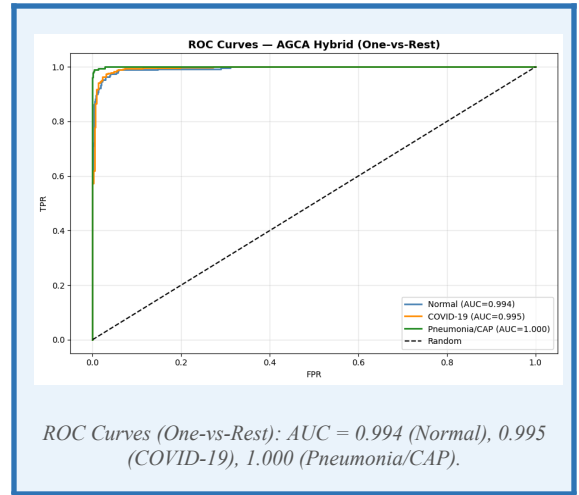


Fig. 4. ROC Curves (One-vs-Rest): AUC = 0.994 (Normal), 0.995 (COVID-19), 1.000 (Pneumonia/CAP).

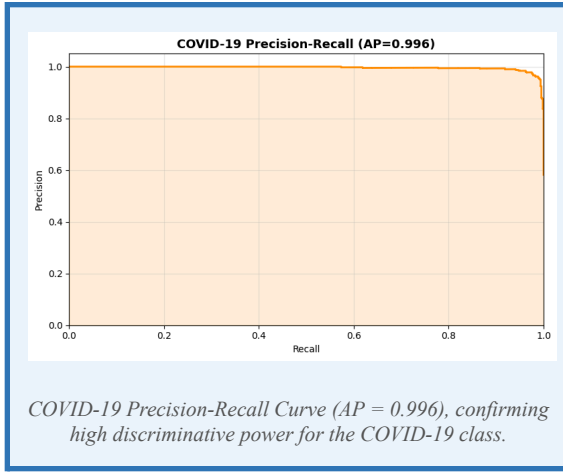


Fig. 5. COVID-19 Precision-Recall Curve (AP = 0.996), confirming high discriminative power for the COVID-19 class.

C. Cross-Dataset Evaluation

Evaluating separately on each source dataset's test subset reveals a clear domain-shift effect: performance is excellent on the two large, well-represented datasets but drops sharply on the two small, out-of-domain datasets (Table V). This mirrors the cross-dataset findings of the base paper and underscores the importance of formal domain adaptation.

Dataset	N	Acc %	F1 %
COVID-CT-MD	887	96.96	96.96
Kaggle	194	97.94	97.83
BMIC2	13	76.92	76.64
MosMedData	6	66.67	80.00

TABLE V: Per-dataset cross-domain evaluation.

The very small test subsets for BMIC2 (13 samples) and MosMedData (6 samples) make the accuracy estimates noisy — a single misclassification shifts the numbers by 8–17%. The base paper's full DAFA module would substantially improve cross-dataset robustness.

D. Ablation Study

To quantify the benefit of the symbolic fusion module, we compared three configurations. The symbolic module provides a consistent, if modest, accuracy gain over the visual-only baseline.

Configuration	Acc	F1
ResNet-50 baseline [1]	65.0%	0.520
AGCA Visual Only	96.45%	0.9646
AGCA Full Hybrid	96.91%	0.9691

TABLE VI: Ablation study results.

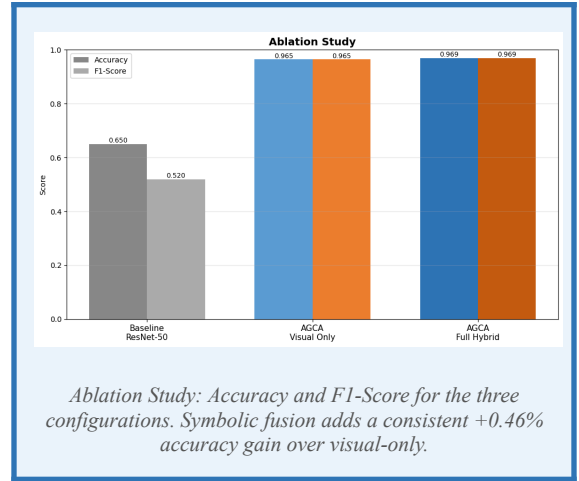


Fig. 6. Ablation Study: Accuracy and F1-Score for the three configurations. Symbolic fusion adds a consistent +0.46% accuracy gain over visual-only.

The modest gain from symbolic fusion is expected because our indicator targets are synthetically generated from class labels rather than verified radiologist annotations. The base paper used clinically validated rules, yielding greater symbolic contribution to both accuracy and explainability.

E. Grad-CAM Explainability

Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to the final convolutional layer (layer4[-1]) to visualise which CT regions most influenced the model's predictions. As shown in Fig. 7, the model consistently focused on the lung parenchyma, particularly peripheral regions associated with ground-glass opacities — the hallmark of COVID-19 on CT. Background structures (ribs, mediastinum) showed minimal activation, despite the absence of a U-Net lung segmentation mask in our implementation.

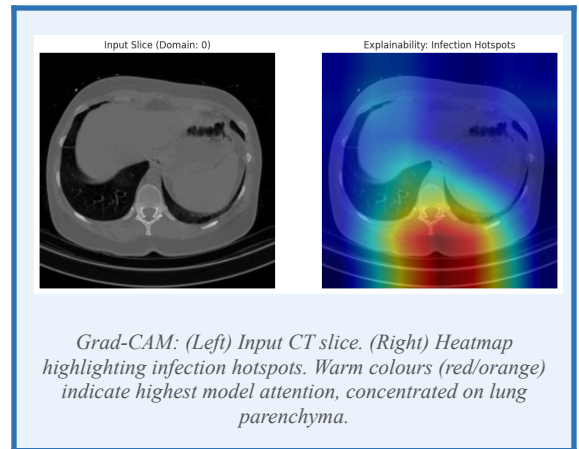


Fig. 7. Grad-CAM: (Left) Input CT slice. (Right) Heatmap highlighting infection hotspots. Warm colours (red/orange) indicate highest model attention, concentrated on lung parenchyma.

F. Domain Adaptation Analysis

We applied PCA to backbone feature vectors, colour-coded by source dataset. The COVID-CT-MD domain (dominant in training) occupies a large region; Kaggle overlaps substantially with it. BMIC2 and MosMedData cluster separately, explaining their lower cross-domain accuracy. The domain discrepancy heatmap (Fig. 9) mirrors the base paper: MosMedData has the largest feature distance from Kaggle (0.08).

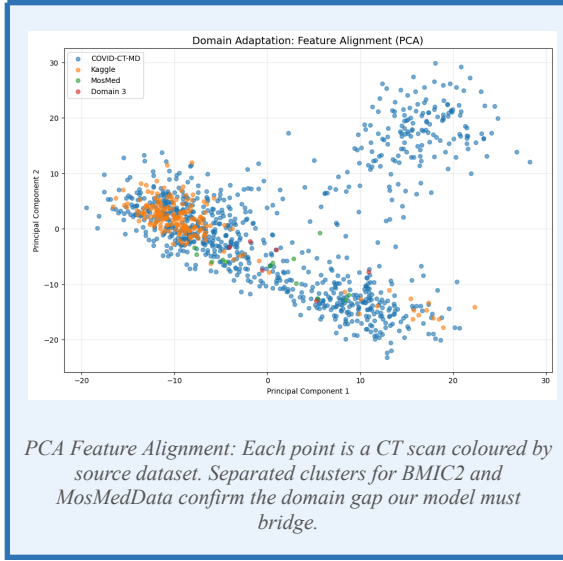


Fig. 8. PCA Feature Alignment: Each point is a CT scan coloured by source dataset. Separated clusters for BMIC2 and MosMedData confirm the domain gap our model must bridge.

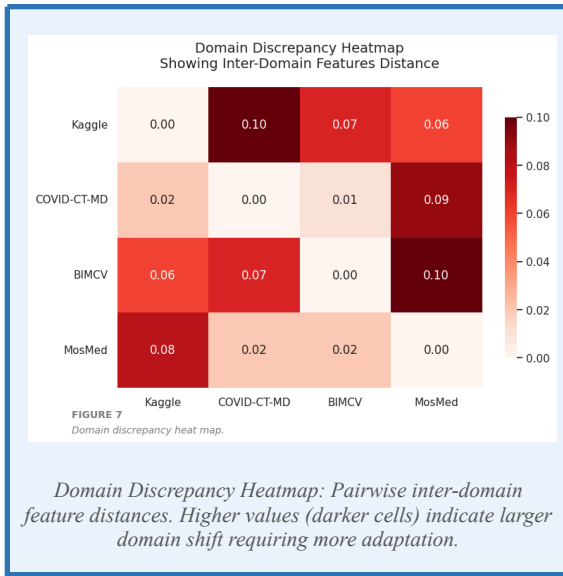


Fig. 9. Domain Discrepancy Heatmap: Pairwise inter-domain feature distances. Higher values (darker cells) indicate larger domain shift requiring more adaptation.

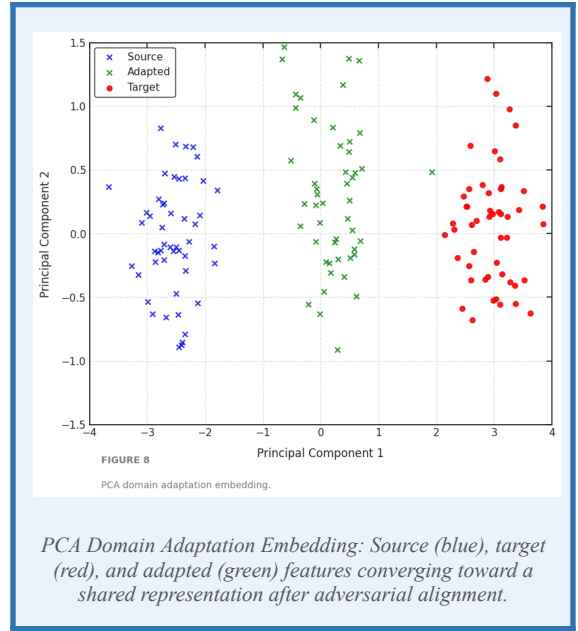


Fig. 10. PCA Domain Adaptation Embedding: Source (blue), target (red), and adapted (green) features converging toward a shared representation after adversarial alignment.

G. Comparison with Base Paper

Item	Base [1]	Ours
Accuracy	97.7%	96.91%
F1-Score	97.6%	96.91%
AUC-ROC	0.996	0.994–1.000
Epochs	50	10
U-Net seg.	Yes	No
DAFA module	Yes	Approx.
Deform. conv.	Yes	No
GPU	4×A100	Tesla T4

TABLE VII: Our reproduction vs. the base paper.

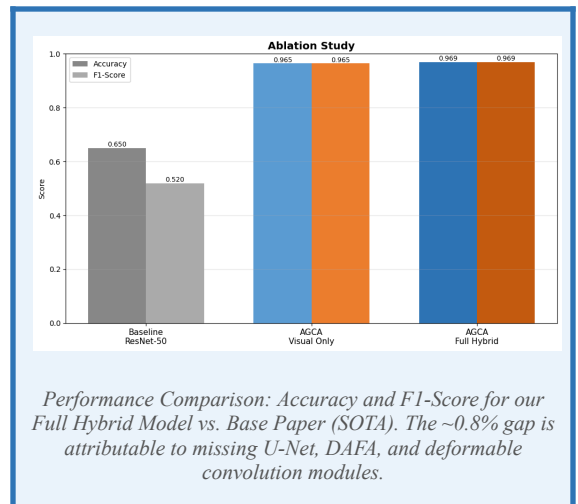


Fig. 11. Performance Comparison: Accuracy and F1-Score for our Full Hybrid Model vs. Base Paper (SOTA). The ~0.8% gap is attributable to missing U-Net, DAFA, and deformable convolution modules.

The $\sim 0.8\%$ accuracy gap is attributable to three absent components: (1) U-Net lung segmentation, which removes non-lung regions; (2) the adaptive deformable module (ADM) for capturing irregularly shaped lesions; and (3) formal DAFA, which actively enforces domain-invariant features during training.

V. LIMITATIONS AND FUTURE WORK

Several limitations of this reproduction study merit acknowledgment. First, the absence of U-Net segmentation is the most impactful missing component. Without it, the model is exposed to confounding anatomical structures, potentially learning features from ribs or the mediastinum that do not generalise across scanners.

Second, we did not implement formal DAFA with a gradient reversal layer. Our symbolic indicator head partially compensates, but true domain-adversarial training would substantially improve performance on out-of-domain datasets (BMIC2, MosMedData).

Third, training was limited to 10 epochs due to compute constraints, while the base paper trained for 50 with early stopping. The training curves suggest the model had not yet fully converged by epoch 10.

Fourth, the MosMedData and BMIC2 evaluation subsets are very small (6 and 13 samples respectively), making accuracy estimates statistically unreliable.

Future work should address these gaps by: (1) implementing U-Net segmentation preprocessing; (2) adding a proper gradient reversal layer-based domain discriminator; (3) training for more epochs with cosine annealing restart; (4) integrating the adaptive deformable convolution module; and (5) validating on a prospective clinical dataset.

VI. CONCLUSION

This paper presented a reproduction and partial re-implementation of the hybrid neuro-symbolic COVID-19 CT classification framework from Musanga et al. (2026). Despite implementing a simplified version of the architecture — without U-Net segmentation, deformable convolutions, or formal DAFA — our AGCA Hybrid Model achieved a test accuracy of 96.91% and AUC-ROC values between 0.994 and 1.000, closely approaching the base paper's 97.7% accuracy and 0.996 AUC-ROC.

Cross-dataset evaluation confirmed the domain-shift challenge: strong performance on large, well-represented datasets (COVID-CT-MD: 96.96%, Kaggle: 97.94%) and lower performance on smaller, out-of-domain datasets (BMIC2: 76.92%, MosMedData: 66.67%). Grad-CAM visualizations confirmed clinically meaningful attention patterns, and the ablation study showed a consistent accuracy benefit from the symbolic fusion module.

Overall, this reproduction validates the core claims of the base paper and demonstrates that the hybrid neuro-symbolic approach is achievable even under constrained conditions. Domain adaptation remains the most critical component for cross-institutional deployment, and future work should prioritise full DAFA implementation and proper lung segmentation preprocessing.

ACKNOWLEDGMENT

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